

## CPSC 292: Introduction to Exploratory Data Analysis

[Lecture in which CLO is introduced](#) | [Assignments](#) | [Projects](#) | [Exams](#)

### Detailed Course Learning Objectives:

At the completion of this course, students should be able to:

1. Understand the basic structure and function of the *R* programming language.
  - 1.1. Understand the *R* Workspace.
    - 1.1.1. Understand how to use the command line. | [L1.4](#) | [A1.5](#) | [P1](#) | [E1](#)
    - 1.1.2. Understand how to use the help function of *R*. | [L1.4](#) | [A1.5](#) | [P1](#) | [E1](#)
    - 1.1.3. Understand the basic syntax of the *R* language. | [L1.4](#) | [A1.5](#) | [P1](#) | [E1-E3](#)
    - 1.1.4. Execute inbuilt mathematical functions to perform calculations in *R*. | [L1.4](#) | [A1.5](#) | [P1](#) | [E1-E3](#)
    - 1.1.5. Learn how to assign variables in *R*'s environment. | [L1.4](#) | [A1.5](#) | [P1](#) | [E1](#)
    - 1.1.6. Understand the basic syntax of functions in *R*. | [L1.4](#) | [A1.5](#) | [P1](#) | [E1,E3](#)
    - 1.1.7. Open, edit and save a script in RStudio's editor. | [L1.4](#) | [A1.5](#) | [P1-P2](#) | [E1-E2](#)
    - 1.1.8. Understand the concept of working directories. | [L1.2, L1.4](#) | [A1.1-1.3](#) | [P1,P2,P4](#) | [E1,E2](#)
    - 1.1.9. Understand the data classes of *R*. | [L1.5](#) | [A1.6](#) | [P1](#) | [E1](#)
    - 1.1.10. Create vectors, arrays, matrices, lists, and data frames. | [L1.5-1.9](#) | [A1.6-1.12](#) | [P1](#) | [E1](#)
    - 1.1.11. Understand vectors and vectorized calculations. | [L1.5-1.9](#) | [A1.7](#) | [P1](#) | [E1-E3](#)
    - 1.1.12. Learn how to index vectors, arrays, matrices, lists, and data frames. | [L1.5-L1.9](#) | [A1.7-1.12](#) | [P1](#) | [E1-E3](#)
    - 1.1.13. List the special values in *R*. | [L1.9](#) | [A1.13](#) | [P1](#) | [E1](#)
    - 1.1.14. Create ordered and unordered factors. | [L1.9](#) | [A1.13](#) | [P1](#) | [E1](#)
    - 1.1.15. Learn to coerce data from one class into another. | [L1.9](#) | [A1.13](#) | [P1](#) | [E1,E2](#)
  - 1.2. Understand the basic principles of computer programming.
    - 1.2.1. Understand the way computers execute commands. | [L1.2, L1.3](#) | [A1.1, A1.2, A1.4](#) | [P1-P4](#) | [E1-E3](#)
    - 1.2.2. Create functions in *R*. | [L3.1, L3.7, L3.8](#) | [P2](#) | [E3](#)
    - 1.2.3. Use functions to reduce repetitive procedures in a script. | [L3.7, L3.8](#) | [P2](#) | [E3](#)
    - 1.2.4. Use functions to automate and standardize the production of a product (e.g. a graph, an analysis). | [L3.7, L3.8](#) | [P2,P4](#) | [E2-E3](#)
    - 1.2.5. Create a function that vectorizes a calculation. | [L3.7, L3.8](#) | [P2](#) | [E3](#)
    - 1.2.6. Understand and successfully execute a **while** loop. | [L3.5](#) | [P2](#) | [E3](#)
    - 1.2.7. Understand and successfully execute conditional **if/else** statements (vectorized and non-vectorized). | [L3.4](#) | [P2](#) | [E3](#)
    - 1.2.8. Understand and successfully execute **repeat** and **for** loops. | [L3.5](#) | [P2](#) | [E3](#)

- 1.2.9. Use a loop to automate a calculation or procedure. | L3.6 | P2,P4 | E3
- 1.2.10. Use a conditional statement to automate a calculation or procedure. | L3.6 | P2 | E3
- 1.3. Understand the basic principles of software design.
  - 1.3.1. Understand why and how code should be documented. | L3.1 | P2,P4 | E3
  - 1.3.2. Create documentation within code and outside of code in the form of a README file. | L3.10 | P2,P4 | E3
  - 1.3.3. Use whitespace effectively to make scripts more readable. | L1.10, L3.6, L3.10 | P2,P4 | E3
  - 1.3.4. Create and maintain R Project files in RStudio. | L3.2, L3.3, L3.10 | P3,P4 | –
  - 1.3.5. Understand the basic principles of refactoring. | L3.6, L3.11 | P2,P4 | E3
  - 1.3.6. Execute code refactoring toward a specific goal (e.g. improving speed, improving readability). | L3.11 | P2,P4 | E3
- 2. Understand and follow best practices in scientific computing.
  - 2.1. Properly organize your work.
    - 2.1.1. Understand how file systems are structured and organized. | L1.2, L3.1 | A1.2 | E1, E3 | P2
    - 2.1.2. Understand how to navigate file systems using a GUI interface. | L1.2 | A1.2 | E1, E3 | P2
    - 2.1.3. Use directories to organize course work. | L1.2 | A1.3 | E1, E3 | P2
    - 2.1.4. Use standard organization to organize an R project. | L3.1 | E3 | P2
  - 2.2. Understand the importance of reproducibility in scientific data analysis.
    - 2.2.1. Create reproducible scripts in *R*. | L1.4, L2.2, L3.1, L3.10 | P2,P4 | E2,E3
    - 2.2.2. Include effective documentation in scripts and projects. | L2.2, L3.1, L3.10 | P2 | E3,E4
    - 2.2.3. Understand the Open Science movement and the role of data repositories in research. | L3.1,L3.10 | P2,P4 | E3
    - 2.2.4. Create and use Notebooks and documents using RMarkdown. | L2.2, L3.10, L4.4, L4.5 | P1,P2,P4 | –
- 3. Independently perform basic data analysis and visualizations in a way that communicates ideas clearly.
  - 3.1. Load, clean, and organize data in *R*. | L2.3 | E2 | P1,P2,P4
    - 3.1.1. Load text, CSV, Excel data files and built-in package data sets. | L2.3 | P1,P2,P4 | E2
    - 3.1.2. Clean, arrange, and transform data sets. | L2.3 | P1,P2,P4 | E2
    - 3.1.3. Learn to index specific values in data sets. | L2.3 | P1,P2,P4 | E2
    - 3.1.4. Create and use data frames from other data types. | L2.3 | P1,P2,P4 | E2
  - 3.2. Learn how to plot quickly using *R*'s base graphics. | L2.2 | P1-P4 | E2

- 3.2.1. Create bar graphs, line, scatter, box, and histogram plots. | L2.2 | P1 | E2
- 3.2.2. Learn when to implement each type of plot above. | L2.2 | P1 | E2
- 3.2.3. Create a plot legend. | L2.2 | P1 | E2
- 3.2.4. Alter the axes, title, and labels of a plot. | L2.2 | P1 | E2
- 3.2.5. Create text labels on a plot. | L2.2 | P1 | E2
- 3.3. Learn the basics of `ggplot2`. | L2.6 | P1-P4 | E2
  - 3.3.1. Understand the basic grammar of graphics used by `ggplot2`. | L2.6 | P1,P2,P4 | E2
  - 3.3.2. Create line, scatter, box, and histogram plots with `ggplot2`. | L2.6 | P1,P2,P4 | E2
  - 3.3.3. Make multi-panel plots. | L2.6 | P1,P2,P4 | E2
  - 3.3.4. Create a plot legend. | L2.6 | P1,P2,P4 | E2
  - 3.3.5. Alter the axes, title, and labels of a plot. | L2.6 | P1,P2,P4 | E2
  - 3.3.6. Create text labels on a plot. | L2.6 | P1,P2,P4 | E2
- 3.4. Execute qualitative and quantitative data analyses. | L2.5 | P1,P2,P4 | E2,E3
  - 3.4.1. Learn the the basic ways data are described. | L2.5 | P1-P4 | E2,E3
  - 3.4.2. Learn the `head`, `tail`, and `summary` commands. | L2.5 | P1-P4 | E2
  - 3.4.3. Calculate the mean, median, quartiles, range, and standard deviation of a data set. | L2.5 | P1-P4 | E2
- 3.5. Think and work independently with code. | L1.10, L2.4, L2.5, L2.6, L3.9, L3.10, L3.11 | P1-P4 | E1-E3
  - 3.5.1. Learn basic Eills in debugging and troubleshooting error messages. | L1.10, L2.4, L3.8 | A1.14 | P1-P4 | E1-E3
  - 3.5.2. Search for effective solutions and tools using online resources. | L2.4, L3.8 | P1-P4 | E1-E3
  - 3.5.3. Learn to accept and provide constructive feedback to peers. | L2.10, L2.11, L2.12 | E2.1, E2 | P1, P2
- 3.6. Understand the basic principles of data visualization and communication.
  - 3.6.1. Understand the basic way our brain processes light stimuli into sight. | L2.7 | P1-P4 | E2
  - 3.6.2. Understand the role of contrast in low-level visual processing. | L2.7, 2.9 | P1-P4 | E2
  - 3.6.3. Understand the organization of the ‘what’ and ‘where’ visual processing systems. | L2.7, 2.8 | P1-P4 | E2
  - 3.6.4. Understand how acuity is focused centrally in vision and drops off as you move into the periphery. | L2.7, L2.8 | P1-P4 | E2
  - 3.6.5. Understand the importance of attention and accessibility to communicating ideas. | L2.8 | P1-P4 | E2
  - 3.6.6. Understand the importance of ethical presentation of data through visualization. | L2.8 | P1 | E2
  - 3.6.7. Understand the principle of the Curse of Knowledge and how it impedes communicating ideas. | L2.9 | P1-P4 | E2

- 3.7. Learn the most effective strategies for written and oral presentation of scientific ideas.
  - 3.7.1. Discuss the reasons for communicating scientific results to a variety of audiences. | [L4.1](#) | [P3-P4](#) | –
  - 3.7.2. Practice reading and understanding scientific papers and grant proposals. | [L4.2](#) | [P4](#) | –
  - 3.7.3. Practice watching and understanding oral presentations (slideshow and poster). | [L4.3](#) | [P3](#) | –